

main.c		Run	Output
<pre>1 #include <stdio.h> 2 #include <stdlib.h> 3 int main() 4 { 5 int request[50]; 6 int n, head, diskSize, direction; 7 int i, j, temp; 8 int totalMovement = 0; 9 printf("Enter number of disk requests: "); 10 scanf("%d", &n); 11 printf("Enter the disk request queue:\n"); 12 for(i = 0; i < n; i++) 13 { 14 scanf("%d", &request[i]); 15 } 16 printf("Enter initial head position: "); 17 scanf("%d", &head); 18 printf("Enter disk size: "); 19 scanf("%d", &diskSize); 20 printf("Enter direction (1 for right, 0 for left): "); 21 scanf("%d", &direction); 22 for(i = 0; i < n - 1; i++) 23 { 24 for(j = i + 1; j < n; j++) 25 { 26 if(request[i] > request[j]) 27 { 28 temp = request[i]; 29 request[i] = request[j]; 30 request[j] = temp; 31 } 32 } 33 } 34 printf("\nSeek Sequence:\n"); 35 printf("%d", head); 36 if(direction == 1) 37 { 38 for(i = 0; i < n; i++) 39 { 40 if(request[i] >= head) 41 { 42 totalMovement += abs(head - request[i]); 43 head = request[i]; 44 printf(" -> %d", head); 45 } 46 } 47 totalMovement += abs(head - (diskSize - 1)); 48 head = diskSize - 1; 49 printf(" -> %d", head); 50 for(i = n - 1; i >= 0; i--) 51 { 52 if(request[i] < head) 53 { 54 totalMovement += abs(head - request[i]); 55 head = request[i];</pre>			<pre>Enter number of disk requests: 6 Enter the disk request queue: 20 35 70 90 10 50 Enter initial head position: 45 Enter disk size: 100 Enter direction (1 for right, 0 for left): 1 Seek Sequence: 45 -> 50 -> 70 -> 90 -> 99 -> 90 -> 70 -> 50 -> 35 -> 20 -> 10 Total Head Movement = 143 cylinders Average Head Movement = 23.83 cylinders === Code Execution Successful ===</pre>